

8_16

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Chapter 8: Heteroskedasticity - Exercise 8.16

(a) OLS Estimation and 95% Confidence Interval

根據 OLS 估計結果，我們關注 kids 變數對旅遊英里數（MILES）的影響。

```
library(POE5Rdata)
library(sandwich)
library(lmtest)
library(dplyr)

data("vacation")
```

```
# OLS model
ols_reg <- lm(miles ~ income + age + kids, data = vacation)
ci_a <- confint(ols_reg, "kids", level = 0.95)
print(ci_a)
```

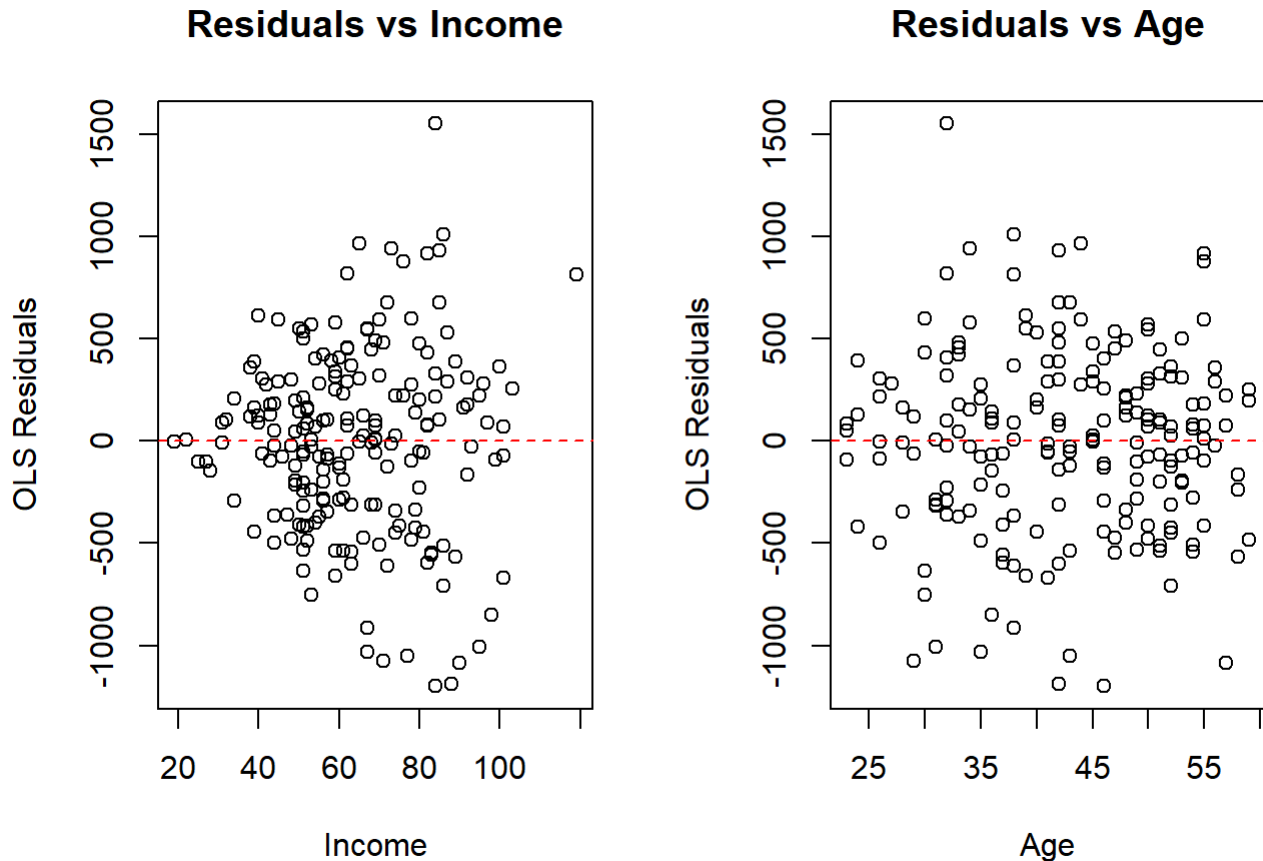
```
##           2.5 %    97.5 %
## kids -135.3298 -28.32302
```

Interpretation: 在維持所得與年齡不變的情況下，多增加一名小孩對於旅遊英里數的邊際影響之 95% 信賴區間如上所示。若此區間不包含 0，則代表小孩人數對旅遊距離具有顯著影響。

(b) Residual Plot Diagnosis

透過殘差圖判斷是否存在異質變異。

```
res <- resid(ols_reg)
par(mfrow=c(1,2))
plot(vacation$income, res, main="Residuals vs Income", xlab="Income", ylab="OLS Residuals")
abline(h=0, col="red", lty = 2)
plot(vacation$age, res, main="Residuals vs Age", xlab="Age", ylab="OLS Residuals")
abline(h=0, col="red", lty = 2)
```



Observation: 觀察 Residuals vs Income 圖，可以發現隨著所得增加，殘差的散佈範圍有逐漸擴大 (Fan-shaped/Spray pattern) 的趨勢。這初步暗示了誤差項可能存在與所得相關的異質變異 (Heteroskedasticity)。

(c) Goldfeld–Quandt (GQ) Test

我們針對所得 (Income) 進行 GQ 檢定，測試變異數是否隨所得增加而上升。Hypotheses: $H_0 : \sigma_i^2 = \sigma^2$ (Homoskedasticity) $H_1 : \sigma_i^2 = \sigma^2 \times INCOME_i^2$ (Heteroskedasticity)

Test Procedure & Results:

```
# Sort by income and split data
vac_sorted <- vacation %>% arrange(income)
low_90 <- vac_sorted[1:90, ]
high_90 <- vac_sorted[111:200, ]

ols_low <- lm(miles ~ income + age + kids, data = low_90)
ols_high <- lm(miles ~ income + age + kids, data = high_90)

# F-statistic calculation
f_stat <- (deviance(ols_high)/86) / (deviance(ols_low)/86)
p_val_gq <- pf(f_stat, 86, 86, lower.tail = FALSE)

cat("GQ Test F-statistic:", f_stat, "\n")
```

```
## GQ Test F-statistic: 3.104061
```

```
cat("P-value:", p_val_gq, "\n")
```

```
## P-value: 1.64001e-07
```

Conclusion: 若 $p\text{-value} < 0.05$ ，我們拒絕 H_0 ，正式確認模型中存在異質變異。

(d) Robust Standard Errors (HC1)

當異質變異存在時，我們使用 White 穩健標準誤差 (HC1) 來修正 OLS 的推論。

```
robust_cov <- vcovHC(ols_reg, type = "HC1")
robust_se <- coeftest(ols_reg, vcov = robust_cov)

# Manual 95% CI calculation
t_crit <- qt(0.975, 196)
kids_coef <- coef(ols_reg)["kids"]
kids_rob_se <- robust_se["kids", "Std. Error"]
ci_d <- c(kids_coef - t_crit * kids_rob_se, kids_coef + t_crit * kids_rob_se)

cat("OLS (Robust) 95% CI for kids:", ci_d, "\n")
```

```
## OLS (Robust) 95% CI for kids: -139.323 -24.32986
```

Comparison: 與 (a) 小題相比，穩健標準誤差考慮了異質變異的影響。若區間變寬，說明原本的 OLS 標準誤差低估了真實的不確定性。

(e) GLS/WLS Estimation and Comparison

最後，我們假設 $\sigma_i^2 = \sigma^2 \times INCOME_i^2$ ，利用加權最小平方法 (WLS) 進行修正。

```
# WLS with weights = 1/income^2
gls_reg <- lm(miles ~ income + age + kids, data = vacation, weights = 1/income^2)

# Conventional GLS CI
ci_e_conv <- confint(gls_reg, "kids", level = 0.95)

# Robust GLS CI (Double protection)
gls_robust_cov <- vcovHC(gls_reg, type = "HC1")
gls_rob_se <- coeftest(gls_reg, vcov = gls_robust_cov)
kids_gls_coef <- coef(gls_reg)["kids"]
kids_gls_rob_se <- gls_rob_se["kids", "Std. Error"]
ci_e_robust <- c(kids_gls_coef - t_crit * kids_gls_rob_se, kids_gls_coef + t_crit * kids_gls_rob_se)

# Summary Output
results <- rbind(ci_a, ci_d, ci_e_conv, ci_e_robust)
rownames(results) <- c("OLS_Conventional", "OLS_Robust", "GLS_Conventional", "GLS_Robust")
print(results)
```

```
##              2.5 %    97.5 %
## OLS_Conventional -135.3298 -28.32302
## OLS_Robust       -139.3230 -24.32986
## GLS_Conventional -119.8945 -33.71808
## GLS_Robust       -121.4134 -32.19919
```

Discussion: 1. Efficiency: GLS 估計量在滿足權重假設下，通常會比 OLS 具備更小的變異數，因此信賴區間通常較窄。 2. Robustness: 使用 Robust GLS 的目的是為了防範我們對變異數形式的假設 ($1/Income^2$) 不完全正確的情況。 3. Consistency: 所有的估計值應該在數值上相近，這顯示了模型的一致性。